

Decision-making UNDER UNCERTAINTY AND RISK

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UNCERTAINTY

Uncertainty is defined as anything that is not known about the outcome of a venture when the decision is made [1].

RISK

Risk is a particular kind of uncertain event which itself, or its consequence, is considered "negative" to someone in a given context [3].

DECISION-MAKING

Decision-making is referred to as the process whereby an individual, group, or organisation reaches conclusions about what future actions to pursue, given a set of objectives and limits on available resources [2].

RESEARCH QUESTION

- To what degree can a decision be justified in flood prediction under high risk and high uncertainty? How do DBN and FL handle uncertainty and risk in such cases, and how does this interpretation affect advised action?

THE PROBLEM

- There is an increase in **extreme weather events**, especially **floods**. Extreme rainfall events due to climate change have increased the risk of flooding in developed areas. Quantifying this heightened risk is vital for protecting life and property, as well as infrastructure planning and design [4].
- Classic flood prediction models attempt to handle uncertainty and risk when assessing decision-making, but are not considered efficient. Hence, this study explores the implementation of two AI models; Discrete Bayesian network and Fuzzy Logic, capable of contending with uncertainty and risk in flood scenarios.

APPROACH

- Discrete Bayesian Network (DBN)**: algorithm associates flood causes and presents how they directly or indirectly influence FloodProbability (refer to the Model structure). DBN was trained by learning parameters, through the usage of Maximum Likelihood and Bayesian Estimation, which aided in presenting the strength between the antecedents and consequences.
- Fuzzy Logic (FL)**: is simpler than DBN, it is an IF-THEN rule based system, that conveys the relationship between inputs and outputs. Utilising the fuzzy inference system, we gather a series of rules to make a decision [5]. This process involves determining the degree of truth in the input data, combines rules through logic operations and generates fuzzy output, which is subsequently converted into an action by defuzzification.

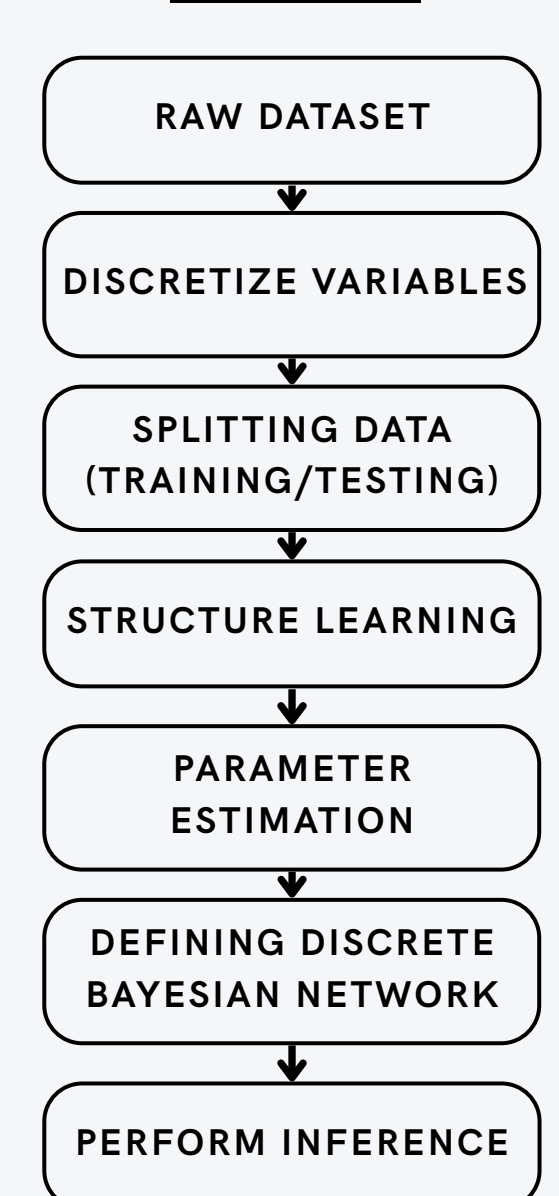
RESULTS

- The bar graph presented on the right-hand side, shows how often each model (DBN & FL) recommend specific actions ("Evacuate!", "Warning", and "No Issue"). It is important to note that this is **not a performance or quantitative comparison**. Instead, this research focuses on a **qualitative evaluation** of how these two methods manage uncertainty and risk, and how this impacts our decision-making procedure.
- DBN recommends "Evacuate!" more frequently than "Warning", presenting a more **conservative** behaviour under the effect of both uncertainty and risk. Meanwhile, Fuzzy Logic, shows an elevated number of "Warning", in comparison to "Evacuate!" and "No Issues", indicating a more **liberal** behaviour than DBN.

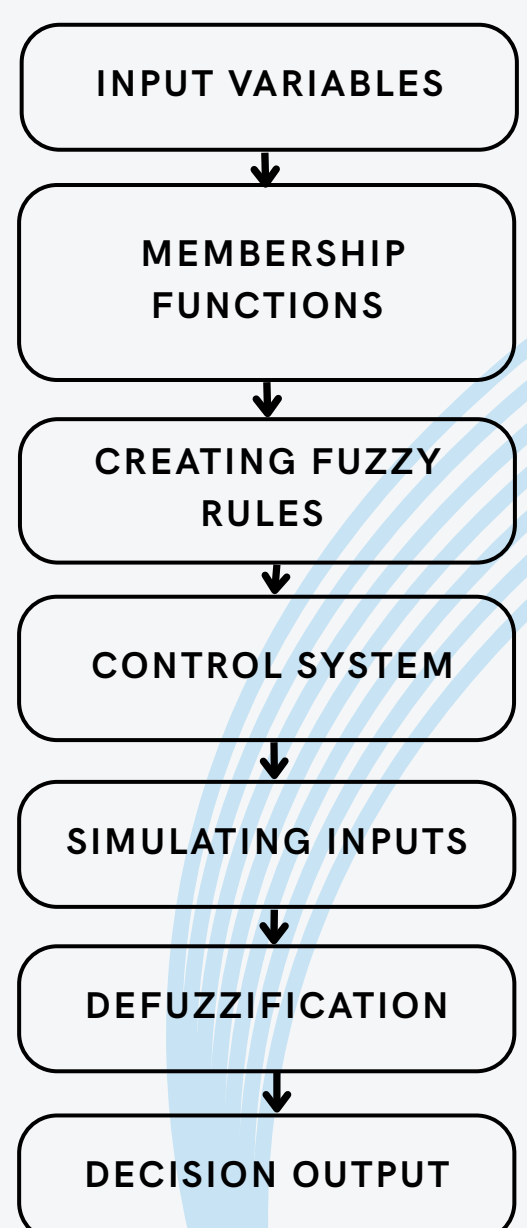
CONCLUSION

- Both Discrete Bayesian Network and Fuzzy logic effectively handle uncertainty and risk in their own way, in the context of floods as an extreme weather event. Through the support of these two models, decision-making can be conducted securely and adequately. Future avenues could possibly involve merging both models.

DISCRETE BAYESIAN NETWORK



FUZZY LOGIC



SHARED RESULT

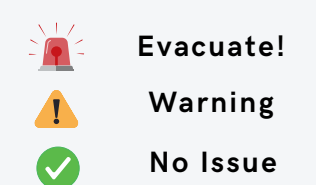


Figure: Step-by-step procedure (DBN vs FL)

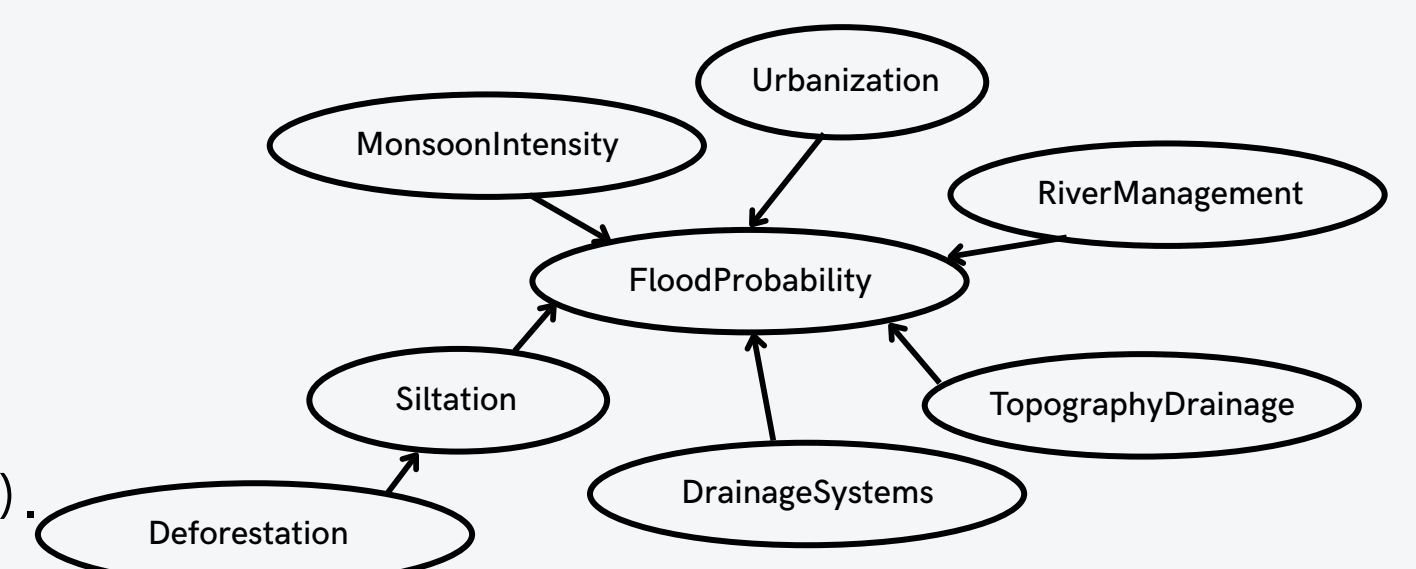


Figure: Model Structure

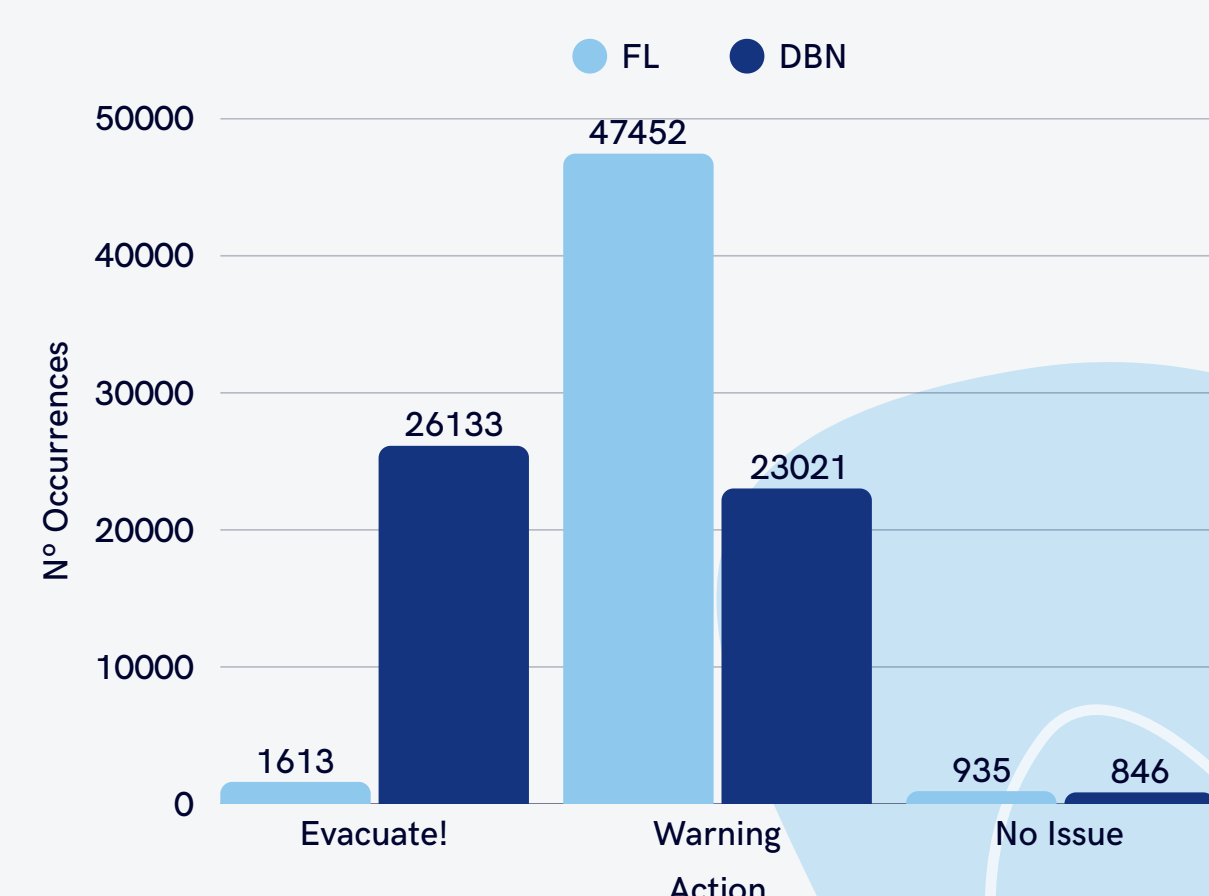


Figure: Model Recommendation Distribution (FL vs. DBN)

REFERENCES

- [1] Townsend, D. M. et al. (2018). Uncertainty, Knowledge Problems, and Entrepreneurial Action. Academy of Management Annals, 12(2), 659-687. <https://doi.org/10.5465/annals.2016.0109>
- [2] S. Lambert, "Decision-making of educational leaders: new insights from the Iowa gambling task," Journal of Work-Applied Management, Feb. 2025, doi: 10.1108/jwam-09-2024-0137.
- [3] S. H. Begg, M. B. Welsh, and R. B. Bratvold, "Uncertainty vs. Variability: What's the Difference and Why is it Important?," SPE Hydrocarbon Economics and Evaluation Symposium, May 2014, doi: 10.2118/169850-ms.
- [4] S. Hettiarachchi, C. Wasko, and A. Sharma, "Increase in flood risk resulting from climate change in a developed urban watershed - the role of storm temporal patterns," Hydrology and Earth System Sciences, vol. 22, no. 3, pp. 2041-2056, Mar. 2018, doi: 10.5194/hess-22-2041-2018.
- [5] K. Raza, "Fuzzy logic based approaches for gene regulatory network inference," Artificial Intelligence in Medicine, vol. 97, pp. 189-203, Dec. 2018, doi: 10.1016/j.artmed.2018.12.004.

